Reading group notes: long-term memory

Reading group of the Ashbury machine learning lab

1 Background

A practical system has to balance the cost of the read gate against the quality of the write policy. We argue that the test-time update deserves as much attention as the replay buffer. Practitioners report that the forgetting rate is the first component to fail on long inputs. Earlier long-term memory methods improved the write policy but left the test-time update unchanged.

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. We expect the replay buffer to interact with the associative loss at larger scales, which we leave to future work. Combining long-term memory with methods that target the write policy is a promising direction. The write policy also makes long-term memory easier to inspect, since its behaviour can be read off per token.

This work studies long-term memory, a method that rethinks the replay buffer. Practitioners report that the retrieval key is the first component to fail on long inputs. Despite steady progress, the interaction between the associative loss and the retrieval key is still poorly understood. A practical system has to balance the cost of the replay buffer against the quality of the memory slot.

2 Methods we read

Unlike a recurrent baseline, long-term memory updates the write policy only when the retrieval key changes. The replay buffer in long-term memory is parameterised by a small linear layer with 128 hidden units. We implement the read gate as a fused kernel, which avoids writing intermediate results to memory. A coefficient controls how strongly the retrieval key depends on the read gate; we set it to 0.7 throughout.

Unlike a memory-free model, long-term memory updates the memory slot only when the memory capacity changes. The read gate in long-term memory is parameterised by a small linear layer with 64 hidden units. A coefficient controls how strongly the read gate depends on the forgetting rate; we set it to 0.5 throughout. To keep training stable, long-term memory normalises the memory slot before it is combined with the replay buffer.

Unlike a sliding-window Transformer, long-term memory updates the segment summary only when the replay buffer changes. A coefficient controls how strongly the retrieval key depends on the associative loss; we set it to 0.6 throughout. The test-time update is initialised so that long-term memory starts training close to a recurrent baseline. To keep training stable, long-term memory normalises the read gate before it is combined with the associative loss.

3 Results we discussed

On PassKey-Synth, long-term memory improves over a sliding-window Transformer by 1.9 points. We train every model for 40B tokens with the same data order and report the mean of three seeds. Ablating the associative loss lowers the LongRecall score by 3.5 points, the largest drop among all components. The gains of long-term memory grow with the input length, as the memory capacity becomes the bottleneck.

With the associative loss frozen, long-term memory still recovers most of its improvement on BookQA. The gains of long-term memory grow with the input length, as the associative loss becomes the bottleneck. We train every model for 70B tokens with the same data order and report the mean of three seeds. Long-term memory scores 87.1% on PassKey-Synth, compared with 72.1% for a sliding-window Transformer.

Long-term memory scores 90.4% on PassKey-Synth, compared with 87.1% for a memory-free model. Ablating the test-time update lowers the LongRecall score by 5.4 points, the largest drop among all components. Throughput is measured with a batch size of 64 on a single accelerator. The gains of long-term memory grow with the input length, as the write policy becomes the bottleneck.

4 Open questions

We expect the retrieval key to interact with the memory slot at larger scales, which we leave to future work. The results suggest that the write policy is the main lever for quality at a fixed budget. The memory slot also makes long-term memory easier to inspect, since its behaviour can be read off per token. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. Combining long-term memory with methods that target the write policy is a promising direction. We expect the forgetting rate to interact with the replay buffer at larger scales, which we leave to future work. A limitation of long-term memory is that the segment summary must be tuned separately for each model size.